

D8 Refraction and Total Internal Reflection

Data: Refractive index of crown glass = 1.51

Refractive index of flint glass = 1.61

Refractive index of water = 1.34

Refractive index of cubic zirconia = 2.16

Refractive index of diamond = 2.42

Take the refractive index of air to be 1.00

Complete the table to show the missing angles.

	Light passing from...		...to	
	Material	Angle of incidence /°	Material	Angle of refraction /°
D8.1	Air	50	Crown glass	(a)
	Flint glass	20	Air	(b)
	Water	63	Flint glass	(c)

D8.2 Calculate the speed of light in:

- water as a fraction of the speed of light in air.
- water as a fraction of the speed in diamond.

D8.3 The critical angle for light passing from opal into water is 67.5° . Calculate the refractive index of opal.

CHAPTER D. WAVES

Answers

D8.1

a) $\arcsin\left(\frac{\sin 50^\circ}{1.51}\right) = 30.485^\circ$

b) $\arcsin(\sin 20^\circ \times 1.61) = 33.412^\circ$

c) $\arcsin\left(\frac{1.34}{1.51} \times \sin 63^\circ\right) = 47.866^\circ$

D8.2

a) $\frac{1}{1.34} = 0.7463$

b) $\frac{2.09}{1.34} = 1.80597$

D8.3 $n_{\text{opal}} \times \sin 67.5^\circ = 1.34 \times \sin 90^\circ = 1.34$
 $n_{\text{opal}} = 1.4504$